

Oluwatobi Melvyn Mayungbo

AI Engineer, LLM and Retrieval Systems

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SUMMARY

Full-stack engineer building AI evaluation and developer tools. Built React review interfaces, FastAPI and PostgreSQL services with Redis, an LLM evaluation harness that separates retrieval from generation, and an MCP server for AI assistants. Ships tested code with Docker and CI.

SKILLS

Engineering and MLOps: Python, React, PostgreSQL, Redis, FastAPI, Docker, CI/CD (GitHub Actions), SQL, DuckDB, GCP Cloud Run, AWS ECS Fargate, Railway, Prefect, dbt-core, MLflow, Evidently, Git, Pytest, Streamlit

AI and LLM Systems: LLM Evaluation and Guardrails, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), Hybrid Search (BM25, Dense Embeddings, RRF), Prompt Engineering, Citation Verification, Hugging Face, ONNX Runtime

Machine Learning: PyTorch, Scikit-learn, LightGBM, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

Data Analysis: Pandas, NumPy, SQL, Anomaly Detection

PROJECTS

Reckon: Payment Reconciliation and Review System | [GitHub](#) | [Live App](#)

Tech: Python, PyTorch, DuckDB, FastAPI, React, Docker, Railway, PostgreSQL

- Designed a human-in-the-loop review interface in React showing match confidence scores, audit trails and suggested resolution actions to streamline exception handling.
- Built and deployed a payment reconciliation system processing multi-source transaction feeds to detect uncollected revenue, duplicate settlements and payout discrepancies, combining deterministic matching rules with ML-assisted discrepancy resolution.

FPL AI Manager | [GitHub](#)

Tech: Python, SQLite, FastAPI, React, Docker, Render

- Built two models that play Fantasy Premier League against the live FPL API: a Manager that carries one squad all season under real transfer, budget and chip rules, and a Best XI optimiser that rebuilds the best legal squad each gameweek.
- Wrote a rules engine, an expected-points model over a multi-gameweek horizon and a leakage-free backfill; cached, rate-limited API client, scheduled jobs and one-command deployment, with 574 tests.

Production-Grade Double-Entry Ledger API | [GitHub](#)

Tech: Python 3.12, FastAPI, PostgreSQL 16, Redis 7, Alembic, Docker

- Built a double-entry ledger and payments API where correctness under concurrency, retries, partial failure and replay is enforced in the database: integer minor units, every transaction summing to zero through a deferred constraint trigger, and an append-only ledger where corrections are reversals.
- Required an Idempotency-Key on every mutating endpoint so a replayed key returns the original response and creates nothing; events go out through a transactional outbox.
- Wrote 138 tests against real PostgreSQL and Redis, including a concurrency test where exactly half of N simultaneous transfers succeed and no balance goes negative.

Fintech Compliance RAG Assistant | [GitHub](#) | [Live App](#)

Tech: Python, Streamlit, BM25, ONNX Runtime, Reciprocal Rank Fusion, FastAPI, Docker, GCP Cloud Run

- Ran retrieval and generation evaluation as separate metrics to localise failure by component; traced a 0.40 refusal-correctness result to a lexical-overlap limitation in the offline stub judge and flagged re-validation against a live provider as the next step instead of overstating readiness.

EXPERIENCE

Machine Learning Instructor, SQI College of ICT

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
 - Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.
- Internal Auditor, NISER** May 2024 to July 2024
- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

EDUCATION

B.Sc. Statistics, University of Ibadan	January 2021 to March 2025
Professional Diploma in Artificial Intelligence, SQI College of ICT	July 2025 to Present

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action